

# Ryan Zomorodi

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## EDUCATION

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### University of Illinois Chicago

*Master of Science, Biostatistics*

- Cumulative GPA: 4.0/4.0

Chicago, IL

*Aug 2023 — May 2025*

### University of Illinois Chicago

*Bachelor of Science, Biological Science*

- Minors: Computer Science & Public Health
- Graduated with Highest Distinction
- Honors College
- Cumulative GPA: 4.0/4.0

Chicago, IL

*Aug 2019 — May 2023*

## WORK EXPERIENCE

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### Research Associate

Children's Environmental Health Initiative

- Performed geospatial analysis of hospital and physician accessibility utilizing two-step floating catchment area methods.
- Developed Rate Stabilizing Toolbox in Python to generate generating reliable, local-level age-standardized measures of chronic disease using the restricted univariate conditional autoregressive (UCAR) bayesian model.
- Generated measures of hurricane exposure for 15 severe storms in order facilitate analysis of hurricane exposure effects on mortality amongst elderly individuals.
- Instructed state, tribal, and local public health professionals on the use of GIS to assess and address chronic health disparities.

Jun 2023 — Present

*Chicago, IL*

### Co-Lead

GeoAdvocates

- Designed, in collaboration with another student, a two-part workshop series curriculum to teach students within multiple classes outlining medical deserts within Chicago and how students can utilize Geographic Information Systems (GIS) to expose Chicago health disparities.
- Presented to several UIC courses (SOC251, PUBH100, US101, SOC100, and PPOL100) and registered student organizations.
- Designed, collected, and analyzed survey data taken from participants before and after each workshop.
- Discussed findings at conferences including the UIC Minority Health Conference and UIC School of Public Health Research and Scholarship Week Lightning Talks.

Dec 2021 — Jun 2024

*Chicago, IL*

### Research Assistant

Department of Rheumatology, University of Illinois Chicago

- Conducted and oversaw experimental techniques such as ELISA and qRT-PCR.
- Independently, recorded and analyzed experimental data using both Excel & R.
- Relayed analyses to principal investigator & discussed findings.
- Implemented bioinformatic analysis of RNA-seq data including differential expression, weighted network gene correlation analysis (WGNCA), Passing Attributes between Networks for Data Assimilation (PANDA), and more.

July 2018 — May 2023

*Chicago, IL*

### Pharmacy Technician

Walgreens

- Utilized electronic health records software to access patient medical records, input insurance, allergy, and health condition records, and resolve patient inquiries.
- Resolved insurance inquiries and resubmitted claims in order to ensure patient prescription adherence.
- Filled, mixed, and dispensed prescriptions for approval by pharmacists.

July 2021 — May 2023

*Chicago, IL*

## PROJECTS

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### Scientific Software

- **Rate Stabilizing Toolbox (RSTbx)** - <https://cehicapacitybuilding.org/RateStabilizingToolbox/>  
Python toolbox for generating reliable, local-level age-standardized measures of chronic disease using the restricted univariate conditional autoregressive (UCAR) bayesian model.
- **RSTr** - <https://cehicapacitybuilding.org/RSTr/>  
R package for generating reliable, local-level age-standardized measures of chronic disease using several different conditional autoregressive bayesian models. This package allows for greater insight into the modeling process than the RSTbx.
- **exposhur** - <https://cehicapacitybuilding.org/exposhur/>  
Internal R package for the Children's Environmental Health Initiative which allows for the calculation of hurricane exposure measures at highly resolved geospatial scales.

### Open Source Software

I use open-source software every day, and I make it a point to give back by maintaining and contributing to open-source projects whenever I can.

- **sasquatch** - <https://docs.ropensci.org/sasquatch/>  
R package that bridges SAS and R. Utilizes the Python package **SASPy** to enable transfer of data between R and SAS, interactive SAS evaluation, basic remote SAS client file management, and rendering of SAS within quarto/rmarkdown documents.
- **legs** - <https://ryanzomorodi.github.io/legs>  
Terminal User Interface (TUI) for viewing R objects like `data.frames`, `matrixs`, `arrays`, `lists`, and `vectors`. Designed to allow quick exploration of deeply nested R objects within a neovim R session.
- **arcpygris** - <https://github.com/ryanzomorodi/arcpygris>  
Python toolbox for downloading Census boundaries into an ArcGIS Pro project. Inspired heavily by the R package **tigris**.
- **socratadata** - <https://ryanzomorodi.github.io/socratadata>  
R package API wrapper that utilizes Rust to parse Socrata datasets into R dataframes. Also features support for Socrata Query Language (SoQL) and the Socrata Discovery API.
- **centr** - <https://ryanzomorodi.github.io/centr>  
R package designed to calculate planar or geodesic weighted mean or median spatial centers.
- **healthatlas** - <https://ryanzomorodi.github.io/healthatlas>  
R package API wrapper for Metopio health atlas data. Allows for easy exploration of health atlas datasets and metadata.

## PUBLICATIONS

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### Variation in One-Year Mortality Following Severe Weather Exposure Among Older Americans by Chronic Health Condition and Sociodemographic Status

*Journal of the American Geriatrics Society*

Feb 2026

[10.1111/jgs.70237](https://doi.org/10.1111/jgs.70237)

### Bayesian Models to Generate Small Area Estimates of Population Health: Tutorial for Using Rate Stabilizing Tools and Their Output

*Journal of Medical Internet Research Public Health Surveillance*

Jan 2026

[10.2196/83498](https://doi.org/10.2196/83498)

### Significance of IL-34 and SDC-1 in the pathogenesis of RA cells and preclinical models

*Clinical Immunology*

May 2023

[10.1016/j.clim.2023.109635](https://doi.org/10.1016/j.clim.2023.109635)

### Inhibition of IRAK4 dysregulates SARS-CoV-2 spike protein-induced macrophage inflammatory and glycolytic reprogramming

*Cellular and Molecular Life Sciences*

May 2022

[10.1007/s00018-022-04329-8](https://doi.org/10.1007/s00018-022-04329-8)

### Metabolic reprogramming of macrophages instigates CCL21-induced arthritis

*Immunology & Cell Biology*

Dec 2021

[10.1111/imcb.12512](https://doi.org/10.1111/imcb.12512)

- Metabolic regulation of RA macrophages is distinct from RA fibroblasts and blockade of glycolysis alleviates inflammatory phenotype in both cell types**  
*Cellular and Molecular Life Sciences* Oct 2021  
[10.1007/s00018-021-03978-5](https://doi.org/10.1007/s00018-021-03978-5)
- Interleukin-34 Reprograms Glycolytic and Osteoclastic Rheumatoid Arthritis Macrophages via Syndecan 1 and Macrophage Colony-Stimulating Factor Receptor**  
*Arthritis & Rheumatology* Sep 2021  
[10.1002/art.41792](https://doi.org/10.1002/art.41792)
- Tofacitinib therapy intercepts macrophage metabolic reprogramming instigated by SARS-CoV-2 Spike protein**  
*European Journal of Immunology* Jun 2021  
[10.1002/eji.202049159](https://doi.org/10.1002/eji.202049159)
- CCL25 and CCR9 is a unique pathway that potentiates pannus formation by remodeling RA macrophages into mature osteoclasts**  
*European Journal of Immunology* Jan 2021  
[10.1002/eji.202048681](https://doi.org/10.1002/eji.202048681)
- IRAK4 inhibition: a promising strategy for treating RA joint inflammation and bone erosion**  
*Cellular & Molecular Immunology* May 2020  
[10.1038/s41423-020-0433-8](https://doi.org/10.1038/s41423-020-0433-8)
- Macrophages are the primary effector cells in IL-7-induced arthritis**  
*Cellular & Molecular Immunology* Jun 2019  
[10.1038/s41423-019-0235-z](https://doi.org/10.1038/s41423-019-0235-z)
- Impact of obesity on autoimmune arthritis and its cardiovascular complications**  
*Autoimmunity Reviews* Jun 2018  
[10.1016/j.autrev.2018.02.007](https://doi.org/10.1016/j.autrev.2018.02.007)

## PRESENTATIONS / WORKSHOPS

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- Learning to use the RSTbx to calculate high quality, local level measures of health status**  
*NACDD Surveillance, Epidemiology, and Evaluation Learning (SEEL)* 12 May 2026
- Working with the Tidycensus Package Mini-Course** [Link](#)  
*GIS Capacity Building Project* 9 — 11 Dec 2025
- Beyond the eye of the storm: why localized, multi-dimensional aspects of severe weather exposure matter for vulnerable, older adult populations**  
*CAFE: Hurricanes & Health: Understanding Risks and Fostering Resilience* 4 Aug 2025
- Enhancing Local Level Surveillance and Data Visualization of Health Outcomes: Introducing a Set of Rate Stabilizing Tools**  
*Council of State and Territorial Epidemiologists (CSTE)* 10 Jun 2025
- Enhancing Data to Action: Approaches for Using Maps to Address Chronic Disease**  
*Council of State and Territorial Epidemiologists (CSTE)* 9 Jun 2025
- Encounter with sasquatch: Using SAS, R, and Quarto together** [Link](#)  
*SAS Open Source Community of Practice* 25 May 2025
- Using GIS to Study Social Determinants of Health with R** [Link](#)  
*Northern Arizona University Center for Health Equity Research* 6 — 7 Jun 2024
- An Introduction to ArcGIS Online Mini-Course**  
*GIS Capacity Building Project* 17 — 24 Oct 2023
- A Gentle Introduction to GIS Mini-Course**  
*GIS Capacity Building Project* 19 — 26 Sep 2023

## RNA-sequencing analysis of Macrophage Hyperactivation

*UIC Undergraduate Research Forum*

18 Apr 2023

## Empowering Community Members to Investigate and Address Chicago's Health Disparities

*American Public Health Association*

9 Nov 2022

## Empowering Community Members to Investigate and Address Chicago's Health Disparities

*UIC Minority Health Conference*

[Link](#)

8 Apr 2022

## Developing Student Workshops to Increase Community Engagement in Addressing Chicago's Health Disparities

*UIC School of Public Health Research and Scholarship Week Lightning Talks*

22 Apr 2022

## TEACHING EXPERIENCE

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### 2026 GIS Training for Surveillance of Heart Disease, Stroke, and Other Chronic Diseases in State Health Departments

*GIS Capacity Building Project*

18 Feb 2026 — 4 Jun 2026

### 2025 GIS Training for Surveillance of Heart Disease, Stroke, and Other Chronic Diseases in State Health Departments

*GIS Capacity Building Project*

25 Mar 2025 — 31 Jul 2025

### Hawaii State Department of Health & University of Hawai'i GIS Capacity Building

*GIS Capacity Building Project*

28 Jan 2025 — 12 Jun 2025

### Advanced GIS Training for Surveillance of Heart Disease, Stroke and Other Chronic Diseases Among Health Department Staff

*GIS Capacity Building Project*

13 Nov 2024 - 29 Jan 2025

### 2024 GIS Training for Surveillance of Heart Disease, Stroke, and Other Chronic Diseases in State Health Departments

*GIS Capacity Building Project*

6 Mar 2024 — 31 Jul 2024

### GIS and Chronic Disease Training for Tribal Members and TEC Staff Fall 2023

*GIS Capacity Building Project*

10 Oct 2023 — 31 Jan 2024

## HONORS

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### Highest Distinction

*UIC Department of Biological Sciences*

2023

### Pasteur Prize Nominee

*UIC Department of Biological Sciences*

2023

### Research Grant

*UIC Honors College*

2022

### Travel Grant

*UIC Honors College*

2022

### Maurice Prize

*UIC Honors College*

2022

### Maurice Prize

*UIC Student Leadership and Civic Engagement*

2022

### Undergraduate Most Outstanding Presentation

*UIC School of Public Health Research Week*

2022

### Chancellor's Student Service Award

*UIC Student Leadership and Civic Engagement*

2022

### President's Volunteer Service Award - Bronze

*UIC Student Leadership and Civic Engagement*

2021

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|-------------------------------------------------------------------------------------------------|------|
| <b>1st place - Division 4</b><br><i>Collegiate Chess League</i>                                 | 2021 |
| <b>Chancellor’s Student Service Award</b><br><i>UIC Student Leadership and Civic Engagement</i> | 2021 |
| <b>Research Grant</b><br><i>UIC Honors College</i>                                              | 2022 |

**PROFESSIONAL MEMBERSHIP**

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|--------------------------------------------------------|-------------|
| <b>Harvard Innovation Lab</b>                          | 2022 — 2024 |
| <b>Clinton Global Initiative University Fellowship</b> | 2023        |